Grounding the Earthquake AI

This experiment aims to validate a transferable metadata standard for fiber optic-based seismic sensing. Such a standard would facilitate a calibrated data repository that can be used for training ML models for seismic prediction.

1. Detecting Earthquakes with Internet Cable

Conventional earthquake detection involves using seismometers on the surface of the earth to pick up seismic waves of a certain magnitude. These waves called p-waves begin 15-30 km below the earth's surface, but they are not the destructive s-waves which travel more slowly, and which, if a city is far enough from the fault, might not arrive for tens of seconds after the p-waves.

In DAS, a device called an interrogator sends ultra-short pulses light along the same fiber optic cable that is used to transmit internet across continents and oceans. The interrogator is attached to a strand of this cable and constantly calculates the amount of light that comes back to it via Rayleigh backscattering. When microscopic stresses deform the cable, the interrogator is able to pinpoint where exactly along the cable the stress was detected, thus allowing the cable to be used as a massive seismic sensor.

From this perspective, a major advantage of DAS for seismic sensing is purely geometric: because submarine cables are that much closer to the fault, they can feel the p-waves faster than a detector on the surface. For example, there is a fault 80–120 km off North America's Pacific coast, from northern California up through British Columbia. The nearest major population centres are about 80 km inland, or 180 km from the fault. This means that a cable sitting 20–50 km from the fault will be able to detect the p-wave 20–25 seconds before a land station, and in such a scenario a fiber-based warning system would give residents double the useful warning window to take precautions (from 10–20 seconds to 20–40 seconds). With these potential gains, is not surprising that there has been considerable excitement about integrating DAS into existing earthquake warning networks across the world.

In Monterey Bay (off California) researchers were able to integrated a submarine cable with the West Coast's existing early warning system called ShakeAlert (Gou et al, 2026). In the Canary Islands (near Spain), another groups was able to integrate two 60-km submarine cable segments into Spain's national seismic network (Cubas Armas et al, 2026). Using these systems, the magnitude of earthquakes has been estimated in less than a second, and their location pinpointed with unparalleled accuracy (where conventional land networks struggle to resolve the origination depth of submarine events).

Regional deployments like these on specific types of cables in specific tectonic settings unquestionably count as a scientific advance on our ability to detect earthquakes quickly, precisely and reliably. But their highly regional nature, highly environment-specific nature also presents a challenge to pooling that data to make inferences.

2. Towards an Earthquake-predicting AI

It is not obvious that the future of earthquake detection is a Machine Learning (ML) model that has learned to use various sources of data to identify patterns that historically anticipate an earthquake. But it would be very curious future in which this is not the case.

To see why, first note that DAS data differ from traditional seismic data in several ways: they are sampled densely along a single axial dimension rather than at disperse points, they are strongly affected by coupling to the ground, and (because readings can be ultrahigh-frequency) they can generate huge data volumes. In a seismometer network, the instrument is standardized and well understood. In fiber sensing, every route segment can behave differently depending on how the cable is buried, attached, bent, or stressed. That makes the same earthquake look different depending on where along the cable one looks, and it makes magnitude estimation and arrival-feature picking much harder than the archetypal "detecting a wiggle". So it is not surprising that ML techniques have already proven "essential for improved [seismic] event classification". (Tomasov et al, 2025)

In fact, the future of earthquake prediction might well be something like a *multimodal anomaly predictor* that integrates satellite and DAS data with other seismic data to identify statistical patterns that crystallize prior to an earthquake. This idea is not as far off as it sounds. There is accumulating evidence that anomalous patterns in the Earth's electromagnetic field (and by extension fluctuations in the atmospheric electron density in the ionosphere) may, in a statistical sense, predict earthquakes. The theory here is still in flux, but we know this because there are a few satellites, currently in Low Earth orbit (about 500km altitude), whose job is to monitor the Earth's magnitude field, and to detect the effects of seismic activity on the ionosphere. The point is, sitting in 2026, it does appear that satellite data can be combined with other ground and atmosphere measurements to "reveal a chain of processes before many mainshocks." (DeSantis et al, 2023)

3. The Bottleneck: Unusable Training Data

But even if the best we can do in terms of earthquake forecasting were to pick up from a DAS stream certain slow-slip events, subtle stress changes or low-magnitude foreshocks that predict seismic events that consistently precede and earthquake, it is obvious that training such a model would require years of continuous, dense, ground-truth-labeled seismic data, ideally spatially dense and covering diverse tectonic environments. We might even that say that if integrating fiber optics into seismic networks is leveraging the geographic advantage to advance the *detection* frontier, then leveraging the Big Data possibilities from pooling all these fiber sensing datasets is how we advance the *prediction* frontier.

In this context, one may find it jarring that no one seems to be preparing this dataset at all. The problem is that each DAS deployment speaks its own dialect. To see why, consider that the resolution of a submarine fiber-optic cable is much finer than the network of conventional seismometer instruments. DAS can detect subtle underwater vibrations, from drilling to glacial ice movements to roving whales. To ensure that earthquake warning systems pick up only the "seismologically interesting signals", various interrogator settings, coupling measurements, thermal regime characteristics, and on-site instrument responses – which collectively we call "metadata" – need to be controlled for. Without a validated standard describing how those properties quantitatively affect the real seismic measurements, recordings from different deployments might be pooled into a shared scientific corpus, but they cannot be compared, merged, or used for global inference. Consider, for instance, PubDAS. PubDAS is a public repo of DAS datasets from diverse experiments that follow the RCN's (Research Coordination Network – more on them later) draft DAS metadata model (Mellors et al. 2022). But it turns out that the standard seismic metadata formats "cannot hold all the acquisition parameters needed for the proper characterization of an experiment" and "obtaining a metadata model that fits all the requirements for DAS experiments is challenging and still open to discussion" so as a result "coupling conditions are generally unknown or poorly constrained" (Spica et al, 2023).

Thus, as it stands today, that body of DAS data, while certainly usable, is not as *cumulatively useful* as it might be. Wildly incompatible data cannot be reconciled without some sort of co-located reference seismometer recording contemporaneously. We can think about a metadata standard as reducing the local variance in the data so that the ML (or human researchers) can pick up the

universal indicators of an imminent earthquake. So while the standard is valuable in its own regard, it is an absolute requisite for ML-driven seismic prediction.

That seismology is sitting on the most capable seismic sensing infrastructure it has ever had, but the data are not being generated in a way that makes it possible to train ML models, is obviously a scientific tragedy. This is probably why, from 2020 to 2025, the NSF funded an initiative called the DAS Research Coordination Network whose mandate included developing a draft metadata standard, creating shared data repositories, and coordinating work across various research groups using DAS in geophysics. Sure enough, the DAS RCN did produce a draft standard, but today it is unclear whether any organization has an active mandate or funding to push the standard to completion. Moreover, a validated standard existing per force does not automatically compel anyone to use it. This is to say, there is an opportunity here to do some solid metascience by promoting data sharing, and generally steering the boat of inter-deployment data interpretability. Or, to put another way, the right intervention likely involves producing a validated standard and simultaneously advocating for its adoption. In the section below we discuss the first of these.

4. The Experiment

This experiment is to validate a transferable metadata standard for fiber optic-based seismic sensing. The hypothesis is that transferable standard for multilateral data exchange is actually achievable. Starting with two countries (i.e., *bilateral* data exchange), we will test whether a DAS calibration methodology validated in one submarine environment can be transferred to a geologically distinct environment, without major retraining. The standard will be validated if the model accurately predicts p-wave arrival-times, magnitudes, hypocenters (epicentres), and does not give many false alarms. Once bilateral data exchange can be validated, the path to a multilateral (global) network becomes much more promising.

The Canary Islands deployment has already validated against co-located instruments and integrated into Spain's IGN national network. Meanwhile, in Monterey Bay cable has already integrated with the ShakeAlert network on the West Coast. While we are not constrained to use these particular deployments, the shape of the final experiment is straightforward: can the Canary Islands pipeline process seismic data from Monterey Bay and produce results consistent with ShakeAlert's ground truth? And can Monterey Bay process the Canary islands data and get results consistent with the official records from the Spanish seismic network? Success means we have validated a useful calibration standard across DAS deployments.

In fact, because the Canary Islands cable sits on a volcanic island shelf whereas Monterey Bay is a continental margin, if the pipeline transfers across, the standard will have demonstrated robustness to the kind of environmental variation that will be encountered once a DAS-enabled network is deployed, e.g., in the Pacific Ocean (Japan, Chile, New Zealand, etc.)

The key outcome would be a validated DAS metadata standard compatible with existing seismic data standards. This standard would be a community artefact, likely produced in collaboration through earthquake research groups like EarthScope and using FDSN infrastructure. The standard would have to be accompanied by an instrument-testing protocol for characterizing any DAS deployment against local ground truth, subsequent deployments across the planet.

Having validated the standard, we will have generated a dataset of DAS recordings from two tectonic environments, anchored to co-located reference instruments, and explicitly designed for ML training pipelines. This dataset could then be the seed for a global, ground-truth-calibrated dataset of fiber, satellite and traditional seismological data recordings.

5. Learning by Doing

We said above that DAS systems have been integrated into operational real-time seismic networks in the Canary Islands and Monterey Bay. This means that in both of those places a sufficiently rich metadata model was used to allow extraction of useful seismic signals from the DAS data. So it would seem almost a foregone conclusion that a transferable standard can be developed. This is certainly not the case: a DAS cable being integration with a seismometer network is not the same thing as a DAS cable being co-located with a reference seismometer. As we said before, the ideal situation is to have a certain number of reference seismometers recording contemporaneous ground-truths along the cable at each locale; absent these, other data sources for establishing ground truth will have to be employed. But until we have actually done it, it remains possible that no way exists to reconcile the two sets of readings in a way that might inform a transferable standard.

Or: consider that one expects the standard to be responsive to certain variables such as covering coupling class, seafloor geology, interrogator hardware, the depth regime, etc. But even after these have all been calibrated against, the resulting standard may yet be so “thin” to be useful to anyone. That is, if the event classification (earthquake magnitude, location, etc) success of a pipeline calibrated using the transferable standard is consistently inferior to one recalibrated “bespokely” for each site, this hardly augers well for the adoption of the “standard” among data stakeholders. In such a case, the scenario will have to be revised to ensure it captures the empirically relevant metadata.

References

Gou, Y., Mittal, H., Lin, Y.-H., Wu, Y.-M., et al. (2026). Integrating fiber-optic seismic arrays into earthquake early warning systems with the dEPIC framework. *Scientific Reports*, 16(1), 960

Cubas Armas, M., Ugalde, A., Monfret, T., Ventosa, S., Rodriguez, T., & Villaseñor, A. (2026). "Integrating Submarine DAS into a Regional Seismic Network for Enhanced Offshore Earthquake Location: A Case Study from the Canary Islands." *Seismological Research Letters*.

Fairweather, D.M., Tamussino, M., Masoudi, A., Feng, Z., Barham, R., Parkin, N., Cornelius, D., Brambilla, G., Curtis, A., & Marra, G. (2024). "Characterisation of the optical response to seismic waves of submarine telecommunications cables with distributed and integrated fibre-optic sensing." *Scientific Reports* 14, 31843.

van den Ende, M., Trabattoni, A., Baillet, M., & Rivet, D. (2025). "An analysis of the dynamic range of Distributed Acoustic Sensing for Earthquake Early Warning." *Seismica* 4.

McGuire, J.J. et al. "Fiber-Optic Sensing for Earthquake Hazards Research, Monitoring, and Early Warning." U.S. Geological Survey.

Spica, Z.J., Ajo-Franklin, J., Beroza, G.C., Biondi, B., Cheng, F., Gaite, B., Luo, B., Martin, E., Shen, J., Thurber, C., Viens, L., Wang, H., Wuestefeld, A., Xiao, H., & Zhu, T. (2023). "PubDAS: A PUBlic Distributed Acoustic Sensing Datasets Repository for Geosciences." *Seismological Research Letters* 94(2A): 983–998.

Mellors R.J., Hodgkinson, K.M., Hui Lai, V., & the DAS Research Coordination Network Data Management Working Group. (2022). |Distributed Acoustic Sensing (DAS) Metadata Model."

Biondi, E., Tepp, G., Yu, E., Saunders, J.K., Yartsev, V., Black, M., Watkins, M., Bhaskaran, A., Bhadha, R., Zhan, Z., & Husker, A.L. (2026). "Real-time processing of distributed acoustic sensing data for earthquake monitoring operations." *Seismological Research Letters*.

De Santis, A., Cianchini, G., Perrone, L., Calcara, M., Campuzano, S.A., D'Arcangelo, S., De Caro, M., Di Mauro, D., Fidani, C., Nardi, A., Orlando, M., Piscini, A., Sabbagh, D., & Soldani, M. (2023). "The new frontier of satellite pre-earthquake signal detection: the cases of Swarm and CSES missions." *XXVIII General Assembly of the International Union of Geodesy and Geophysics (IUGG)*, Berlin

Tomasov, A., Zaviska, P., Dejdar, P. et al. "Comprehensive Dataset for Event Classification Using Distributed Acoustic Sensing (DAS) Systems." *Scientific Data* 12, 793 (2025).